AoU Data Types — Inventory and Investment Recommendations

2026 Science Committee Working Group, Scientific Roadmap Refresh · Co-leads: Smoller & Haendel · Draft 2026-06-02

This catalog consolidates the data-type evidence scattered across the Phase-4 recommendations, IC-alignment matrix, peer-biobank comparison, and Lina Sulieman's prior workspace mapping (DRC 2023). It is organized by the AoU Scientific Roadmap's "Core Data" axis (EHR, surveys, physical measures, digital health, genomes, assays) plus a derived-data layer. Effort symbols: S small (existing tooling), M moderate (linkage / curation), L large (new collection / governance / infrastructure).

CURRENTLY IN AOU (BASELINE)

Data type	What's available	Coverage / scale	Status
EHR (OMOP)	Conditions, procedures, medications, labs, vitals; ~95% of cohort linked	~315k participants	live
Surveys (PPI)	Basics, lifestyle, healthcare utilization, SDOH, family/personal history	Full cohort	live
Physical measures	Height, weight, BP, heart rate, hip/waist at enrollment	Full cohort	live
Wearables	Fitbit (step, HR, sleep stage), some Garmin; daily-level streams	~15k participants (opt-in)	live
Genomes	Whole-genome sequencing, short-read; v8 expanded	~245k participants	live
Baseline assays	CBC, basic lipids, HbA1c on participants providing biospecimens	Limited	partial

RECOMMENDED FOR NEW INVESTMENT — RAW DATA

Data type	Evidence + peer comparison	Anchor ICs	Effort
Plasma proteomics (Olink / SomaScan)	Largest peer-comparative gap: UKB 12.2% post-2023, AoU 1.8%. UKB-PPP and FinnGen Olink releases drive the gap. Multi-omics-beyond-genomics hot area (HA1).	NHLBI, NIDDK, NIA, NCI	L
Metabolomics (NMR / untargeted MS)	UKB has NMR metabolomics on ~120k. Paired with proteomics under multi-omics; broad cardiometabolic + nutritional epi use.	NIDDK, NHLBI, NCCIH	L
Stool biospecimens → 16S + shotgun metagenomics	AoU 0.8% vs peer median 11.1% (UKB 9.8%, FinnGen 14%, CKB 11.1%). Field-wide growth, not single-peer artifact. NIDDK gut-microbiome strategic priority.	NIDDK, NCCIH, NCI	L
Imaging release — modality-prioritized: (i) EHR-linked CXR/CT via DICOM, (ii) retinal/fundus (OCT), (iii) prospective MRI subset	AoU 8 imaging pubs total (HA3 thin coverage); UKB 8.1% structurally. Data-supply not demand problem. WG-3 Imaging owns staging plan; WG-1 frames the ask.	NCI, NEI, NIBIB, NHLBI, NINDS	L
Environmental exposure linkage — PM2.5, ozone, NO_2, NDVI, heat, water quality, EJI/EJScreen via residential history	Single largest demand-supply gap: NIEHS 6 strategic priorities vs AoU 7 substantive pubs. Residential history already collected — linkage layer needed, not new data.	NIEHS, NCI, NHLBI, NIMHD	M
Advanced physiologic sensors beyond Fitbit (CGM, BP cuff, ECG patch)	CGM is a 2024 standard-of-care for diabetes; emerging cohort use. Builds on existing wearables infrastructure.	NIDDK, NHLBI	L

RECOMMENDED FOR NEW INVESTMENT — DERIVED / PROCESSED DATA

Data type	Evidence	Anchor ICs / Lead	Effort
NLP-derived phenotypes from clinical notes (CLAMP & successors)	Smoller subgroup 2024 prioritization 3.40; notes already in EHR layer but unprocessed. Pairs with phecode workflow.	NLM, disease-specific ICs	M
Polygenic risk scores (validated, multi-ancestry)	Partial in genomic v8; expand to all major outcomes with cross-validated PRS-CSx. AoU's diverse cohort is the right setting for ancestry-fair PRS.	NHGRI, disease ICs	S
Phecode v8 + Workbench integration	Smoller 2024 prioritization 3.80; v7 done, v8 needed, "needs incorporated/integrated into workbench."	NLM	S
Validated computational phenotype repository (PheKB-style)	Smoller 2024 prioritization 3.80; eMERGE/PheKB algorithm library. Maps to Q4 (new tools needed) of Roadmap 2.0.	NLM, NCATS	M

Data type	Evidence	Anchor ICs / Lead	Effort
Foundation-model embeddings (patient EHR-trajectory dense vectors, pre-computed)	WG-4 AI rec 3.50; 31 AI/foundation-model pubs already in AoU corpus. Lowers barrier for downstream researchers.	NLM; co-funded NHLBI/NCI for use cases	M
Imaging-derived radiomics features	Gated on imaging release. WG-3 owns. Phecode-radiomics crosswalk (WG-3 rec 3.88).	NIBIB, NCI	S*
Ascertainment-bias-adjusted variables	Smoller 2024 prioritization 3.00; inverse-probability weights vs external population reference (e.g., NHANES). Smoller subgroup explicitly flagged.	NLM, NIMHD	M

CONSIDERED AND DEPRIORITIZED

- **Cerebrospinal fluid biomarkers / amyloid-PET** — very high participant burden, low throughput. Defer pending an explicit ADRD-cohort strategy (peer comparison: AoU 4× under-indexed on aging/ADRD but not yet positioned to compete with ADNI / UKB neuroimaging).
- **Single-cell omics on cohort scale** — not technically feasible at AoU scale in this Roadmap horizon.
- **Dermatology photo capture** — despite 14% derm corpus share (peer index 7.8×), known to be a 1–2-group artifact; not a strategic priority.
- **Cross-cohort individual-data joins** (Smoller 2024 prioritization 3.00) — flagged as compelling but with substantial data-use-agreement and IRB-harmonization hurdles; defer pending a Translational-WG (WG-2) governance framework.

ALREADY IN FLIGHT (NO NEW RECOMMENDATION NEEDED)

PheWAS×GWAS infrastructure (Smoller 4.00, addressed by the *All by All* initiative); ancestry-tagged genomic v8 release; cohort and dataset builders; basic Workbench tooling.

Cross-references. Full evidence for each line item lives in `phase4-recommendations/`, `aou-ic-alignment/gap_analysis.md`, `peer-biobanks/emerging_themes.md`, and `aou-roadmap-refresh/temporal_trends.md`. *Imaging-radiomics features are S effort *once imaging is released* (L); the gate is the imaging release itself.